

CONFLICT TYPES

Conflict Type I = Historical Shift (since 2012–2014)

Conflict Type II = Estimation

Conflict Type III = Digitization Error

CLASSIFICATION

Minor = Low amount of farmland is affected

Moderate = Low amount of households/businesses or moderate amount of farmland is affected

Major = Moderate to High amount of households/businesses or high amount of farmland is affected

DISCREPANCY POINTS East of the Cagayan River

CRS = PRS92 (EPSG:4683)

* = Possibly Incorrect

^ = City Outskirts

Northeast Cauayan[^]

16.9717555,121.8281417

16.960770,121.831370

16.942221,121.842630

16.9181125,121.8613251

16.9056547,121.9017108

16.9053544,121.8928313

- Conflict Type I
- Minor Discrepancy

San Luis—Villa Luna Border

16.9732291,121.8003050

- Conflict Type I
- Minor Discrepancy

San Luis—Union Border

16.9687461,121.8124653

- Conflict Type I
- Minor Discrepancy

Union—Villa Luna Border

16.9630304,121.8049590

- Conflict Type I
- Conflict Type II

- Minor Discrepancy

Mabantad—Nagcampegan Border

16.9668756,121.7869454

16.9620635,121.7905526

- Conflict Type II
- Minor Discrepancy

C. Chica—C. Grande Border

16.9528261,121.7897103

- Conflict Type III
- Minor Discrepancy

Catalina—C. Bacareño Border

16.9514149,121.7997035

- Conflict Type II^{*}
- Minor Discrepancy

C. Grande Quadripoint

16.950283,121.787867

- Conflict Type III
- Minor Discrepancy

Andarayan—C. Bacareño Border

16.953434,121.802493

- Conflict Type II

- Minor Discrepancy

San Pablo—Villa Luna Border

16.943762,121.816003

- Conflict Type II
- Minor Discrepancy

Cabugao—San Luis Border

16.9184202,121.8510725

16.9186824,121.8562462

- Conflict Type I
- Conflict Type III
- Moderate Discrepancy

Cassap Fuera—San Pablo Border

16.9210392,121.8178324

16.929158,121.813007

- Conflict Type II
- Minor Discrepancy

Cassap Fuera—Guayabal Border

16.924751,121.813264

- Conflict Type II
- Minor Discrepancy

Cassap Fuera—B. Norte Border

16.921103,121.814419

- Conflict Type II
- Minor Discrepancy

B. Norte—San Pablo Border

16.9190185,121.8149925

- Conflict Type II
- Minor Discrepancy

Baringin Sur—San Pablo Border

16.916480,121.815985

- Conflict Type II
- Minor Discrepancy

Dabburab—San Pablo Border

16.9097616,121.8176170

- Conflict Type II
- Minor Discrepancy

San Pablo—Sta. Luciana Border

16.904552,121.820596

- Conflict Type II
- Minor Discrepancy

Casalatan—Baculod Border

16.8795699,121.8513458

16.8762670,121.8890338

16.8704162,121.9028156

16.8688505,121.9062911

- Conflict Type I
- Minor Discrepancy

Baculod—Maligaya Border

16.8642015,121.8909258

16.8585955,121.8762539

16.8573174,121.8733191

16.8591602,121.8689159

16.8577336,121.8652887

- Conflict Type I
- Conflict Type II
- Conflict Type III*
- Minor Discrepancy

Baculod—Dianao Border

16.8586875,121.8578545

16.8598631,121.8534199

- Conflict Type I
- Conflict Type III
- Minor Discrepancy

Dianao—San Pablo Border

16.8693766,121.8357567

16.8660464,121.8358571

- Conflict Type I
- Minor Discrepancy

Gappal—San Pablo Border

16.8659711,121.8329872

- Conflict Type I
- Minor Discrepancy

Dianao—Gappal Border

16.8622603,121.8392626

- Conflict Type II
- Minor Discrepancy

Gappal—Manaoag Border

16.8130319,121.8231850

- Conflict Type II
- Minor Discrepancy

Angadanan—Cauayan Border[^]

16.809944,121.828565 to 16.825658,121.908706

- Conflict Type II
- **Major** Discrepancy

Dianao—Maligaya Border

16.8527259,121.8677027

16.8490359,121.8657740

- Conflict Type II
- Minor Discrepancy

Buyon—Maligaya Border

16.8432751,121.8777977

16.8327680,121.8969858

16.8310193,121.9055705

- Conflict Type I
- Conflict Type II
- Minor Discrepancy

Maligaya—Villa Concepcion Border

16.8462497,121.9096453

- Conflict Type II
- Minor Discrepancy

Maligaya Elem. School—Annex[^]

16.8757733,121.9232713

- Conflict Type III
- **Major** Discrepancy

Casalatan—Sinippil Border

16.8958128,121.8669308

- Conflict Type II
- Minor Discrepancy

Casalatan—Disimuray Border

16.8889705,121.8787181

16.8829001,121.8756139

- Conflict Type I
- Conflict Type III
- Minor Discrepancy

Buyon—Dianao Border

16.8373585,121.8699493^{*}

16.8308106,121.8633214

16.8237184,121.8585280

- Conflict Type I
- Minor Discrepancy

Linglingay—Manaoag Border

16.8222112,121.8536374^{*}

16.8202533,121.8519451

16.8093508,121.8507737

16.8051045,121.8514766

- Conflict Type I
- Conflict Type II
- Minor Discrepancy

Buyon—Linglingay Border

16.8109552,121.8714951

- Conflict Type I
- Minor Discrepancy

Sta. Maria—Villa Concepcion Border

16.8430121,121.9220664

16.8431020,121.9370479

16.8452273,121.9515315

- Conflict Type II
- Conflict Type III
- Minor Discrepancy

Southeast Cauayan and Villaflor[^]

16.8827207,121.9355753 to 16.7776120,121.9389598

16.747071,121.948038

- Conflict Type II
- **Major** Discrepancy

De Vera—Villa Concepcion Border

16.8270537,121.9312620

- Conflict Type I
- Minor Discrepancy

Rogus—Villa Concepcion Border

16.8277106,121.9460050

16.8259283,121.9433233

- Conflict Type I
- Minor Discrepancy

De Vera—Rogus Border

16.8253693,121.9431225

16.8125006,121.9424448

16.8056311,121.9417273

16.8022811,121.9428014

16.8001537,121.9395696

16.7952045,121.9379777

16.7917928,121.9393918

16.7899395,121.9375719

16.7868101,121.9371452

16.7857851,121.9390153

16.7835846,121.9406887

16.7820534,121.9423119

- Conflict Type I
- Conflict Type II
- Moderate Discrepancy

Cabugao—Casalatan Border

16.8967645,121.8455903

- Conflict Type I
- Minor Discrepancy

DISCREPANCY POINTS West of the Cagayan River

CRS = PRS92 (EPSG:4683)

* = Possibly Incorrect

^ = City Outskirts

Northwest Cauayan[^]

16.9726466,121.7797739

16.9742155,121.7584333 to

16.9573033,121.7535051

16.9521282,121.7519739

16.9515195,121.7441589

16.9440894,121.7328318

16.9301580,121.7241341

16.9282377,121.7148089

- Conflict Type I
- Conflict Type III
- **Major** Discrepancy

Marabulig I—Marabulig II Border

16.9193643,121.7245106 to 16.9126287,121.7249687

- Conflict Type II
- Minor Discrepancy

Marabulig I—Nagrumbuan Border

16.9126287,121.7249687

- Conflict Type II
- Minor Discrepancy

Southwest Cauayan[^]

16.8851843,121.6559561

16.8716399,121.6795871 to 16.8622582,121.7097237

16.8584553,121.7393498

- Conflict Type I
- Conflict Type II
- **Major** Discrepancy

Naganacan—Sillawit Border

16.8707886,121.7412220

- Conflict Type II
- Minor Discrepancy

District I—Labinab Border

16.9315323,121.7730194 to 16.9191583,121.7703043

- Conflict Type I
- Minor Discrepancy

District III—Labinab Border

16.9314591,121.7750621 to 16.9362556,121.7782856

- Conflict Type I
- Minor Discrepancy

Cabaruan—San Fermin Border

16.9561905,121.7577817 to 16.9507601,121.7579742

16.9454512,121.7654377 to 16.9426063,121.7677010

- Conflict Type III
- **Major** Discrepancy

Cabaruan—District II Border

16.9430686,121.7701547

- Conflict Type II
- Moderate Discrepancy

Cabaruan—Turayong Border

16.9434001,121.7736930

16.9443394,121.7747807

16.9484958,121.7736240

- Conflict Type I
- Conflict Type II
- Conflict Type III
- Moderate Discrepancy

Alicaocao—Cabaruan Border

16.9488263,121.7736993 to

16.9560597,121.7755945

- Conflict Type I
- Conflict Type II
- Moderate Discrepancy

Buena Suerte—Culalabat Border

16.9240317,121.7951915

- Conflict Type II
- Minor Discrepancy

Faustino—Nunngungan II Border

16.8849155,121.7681084

- Conflict Type III
- **Major** Discrepancy

Central South Cauayan[^]

16.8541425,121.7679610

TOTALS

Discrepancies: 60

Major Discrepancies: 7

Moderate Discrepancies: 8

Minor Discrepancies: 45

West Discrepancies: 19

East Discrepancies: 41

Outskirts Discrepancies: 6

16.8385513,121.8026059 to 16.8338134,121.8107764

- Conflict Type II
- Conflict Type III
- Moderate Discrepancy

Amobocan—San Antonio Border

16.8639667,121.8087181

16.8637125,121.8029981

- Conflict Type II
- Minor Discrepancy

San Antonio—Sta. Luciana Border

16.8809605,121.8091135

16.8831945,121.8080498

- Conflict Type I
- Minor Discrepancy

Dabburab—San Francisco Border

16.8996737,121.8011719

16.9009727,121.7934250

- Conflict Type II
- Minor Discrepancy

B. Norte—Baringin Sur Border

16.9174174,121.8091542

- Conflict Type II
- Moderate Discrepancy

Marabulig II—Minante I Border

16.9129017,121.7427155

- Conflict Type III
- Moderate Discrepancy

Major Discrepancies' Affected House-holds: 200-500

NOTES — Extra Information

This project is not meant to criticize the City Planning and Development Office

- Most minor discrepancies (especially at the East) do not require ground verification, and only needs digital work by comparing the Assessor's Tax Map with a satellite imagery map.
- For maximal accuracy in verification, we need the Assessor's Tax Map to map out where the ends of boundaries would be, and likely check the neighboring municipalities' Tax Maps (i.e. Luna, Alicia, Naguilian, Angadanan, San Guillermo, Benito Soliven, and possibly Reina Mercedes and San Mariano) if it is at that perimeter.
- We will also likely interview the owners of the households (or strategic houses if the place is dense) involved with the dispute for maximal accuracy.
- Some areas such as the Cabugao—San Luis Border around and above 16.9194020,121.8510666 (see Fig.1) are disconnected to their alleged LGU in terms of access, so they are assumed to be part of the most likely LGU until verified.



- All of the boundaries may require to be redrawn, due to GIS shifts, hurried mapping, and other reasons, especially at remote areas at the East where most are poorly drawn, such as 16.8215267,121.9414260 on the De Vera—Rogus Border (see Fig.2).
- Villaflor is shown to be an exclave (see Fig.3). This may be likely due to online sources from San Guillermo, and the existence of Calaoagan, San Guillermo in between, though the boundary is highly estimated. It does roughly estimate of a ridge. Despite this, we may check if it is actually an exclave or not.
- Despite the disputes, evidence shows that the affected properties show no problems, therefore this is likely a digital problem.
- The Maligaya Elementary School—Annex at 16.8757796,121.9233801 (see Fig.4), which is as far as ~500m, is likely to involve as many as 60 households due to its proximity to the



FIGURE 2

Clear = De Vera

Green = Rogus

Shows a poorly drawn boundary that has less nodes than the average amount of nodes in other areas.

- Floor Center, which is likely in District I, is said to be in District II according to its own Facebook page, which is mentioned multiple times that it is in District II. It may be a business permit typo, due to the proximity of the business to District II.

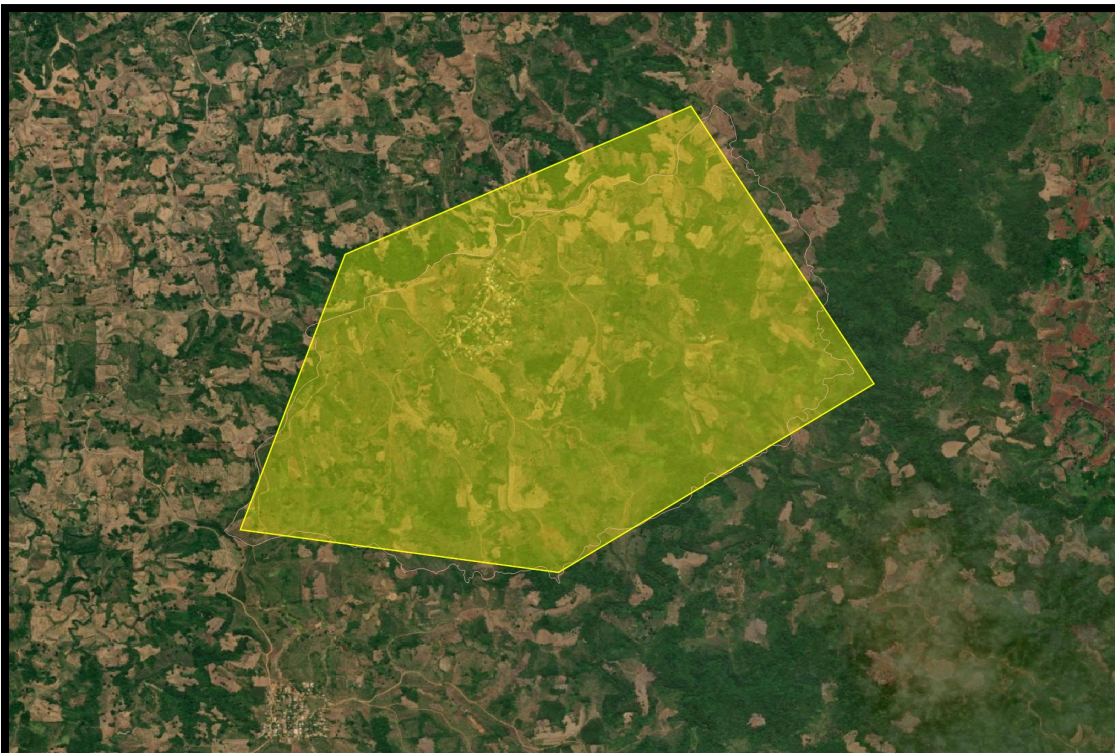


FIGURE 3

Yellow = Villaflor

Red Outline = Ridges

Villaflor is highly estimated and shown to be an exclave surrounded by San Guillermo.



FIGURE 4

Pink = Maligaya

Blue = Santa Maria

Shows the Maligaya School as well the nearby households that are likely involved in the discrepancy.

- BBMs and MBMs (Muhons) that are likely to be at 16.9437955,121.7314073 that also includes the former 'Welcome to Brgy. San Fermin' sign have been demolished due to the construction of the Cauayan City—Cabatuan Diversion Road.
- The Cabaruan—San Fermin Border dispute at 16.9531814,121.7574795 (see Fig.5) also includes a local Child Development Center owned by San Fermin but shown to be in Cabaruan. The neighborhood, known as Purok 8, west of the main road in this area is likely to be part of San Fermin.
- The width of the Angadanan—Cauayan Border dispute may go as far deep as ~300m, from evidence of estimation from barangay boundaries, such as the Gappal—Manaoag Border at 16.812985,121.823241 (see Fig.6).



FIGURE 5

Yellow = San Fermin

Clear = Cabaruan

The whole neighborhood west of the main road going North-South is likely under San Fermin due to development evidence.



FIGURE 6

Purple = Gappal

Red = Manaoag

Shows the border ending with disregard to property lines. This is likely due to the larger Angadanan-Cauayan Border dispute. The nearby household that is also shown may also be affected with this.

- A small neighborhood at 16.9744425,121.7587136 (see Fig.7) is shown to be in Luna, despite only having connections to Cauayan. This is likely under Cauayan due to Luna not connecting to this neighborhood despite having no obstructions to build a connection there. This neighborhood may have also experienced stagnant growth due to this.



FIGURE 7

Peach = Turayong (Cauayan)

Neighborhood is shown to be in Luna despite roadway problems. It may also have experienced stagnant development.

PROJECT ROADMAP

PHASE I

- Obtain the Tax Map from the Assessor's, and obtain the Tax Map from the neighboring municipalities that are involved in the discrepancies upon the perimeter.

PHASE II

- Compare the Tax Maps if on the perimeter, and fix Minor discrepancies that do not require Ground Verification.

PHASE III

- Travel to discrepancies that require Ground Verification to find muhons, and interview owners of the affected properties to compile and gather data to be digitized.

PHASE IV

- Digitize the data and redraw all of the boundaries using the best satellite imagery to be more accurate (e.g. Google Imagery, NAMRIA, Bing).

PHASE V

- Compare our final map to the City Planning and Development Office's (CPDO) map (once they have finished) to create an ultra-accurate map.

PHASE VI

- Present to the Sangguniang Panlungsod (SP) and await their authorization of the map to then present the final perimeter line to the neighboring municipalities involved.

PHASE VII

- Present and await the authorization of the map to the MPDOs, the Assessors, and the Sangguniang Bayans (SB) of the involved municipalities.

PHASE VIII

- Certify the map for legal use and import the data to OpenStreetMap (OSM) via the signed Data License.

PHASE IX

- Establish a feedback loop with the Assessor's and the CPDO notify any property or boundary changes at the boundaries to be able to be fixed. The boundaries will also be monitored for any other shifts in satellite imagery to be fixed.